

POWERING SATELLITES

Early satellites relied on batteries that drained rapidly in the cold of space, giving them a limited lifetime. The US's second satellite, Vanguard 1, was launched in 1958 and tested the idea of using solar panels to generate electricity from the sun. Today, almost all satellites rely on solar power.

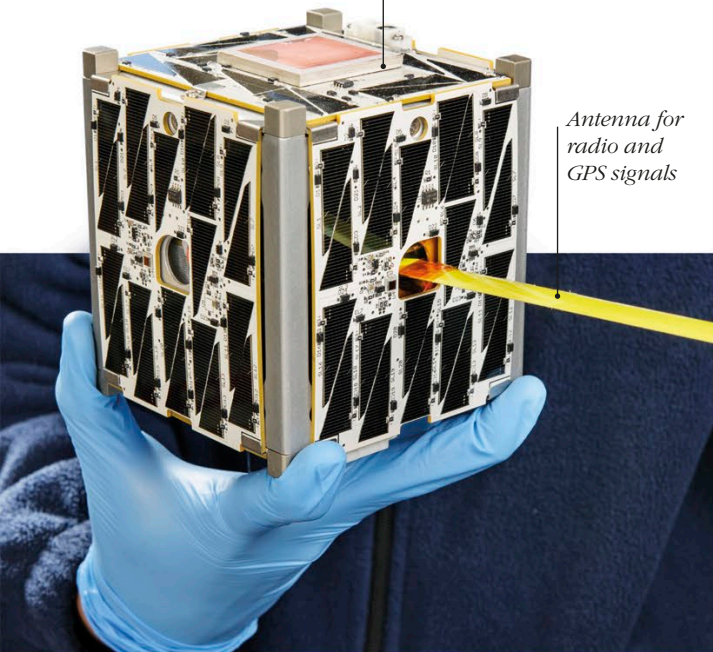


Camera gathers information about Earth from above in an activity known as "remote sensing"

Solar "wing" is a 76-sq-ft (7.1-sq-m) array of three solar panels, generating up to 2,300 watts of power

NASA's Phonesat 2.5 uses smartphone components powered by solar cells.

Illustration of a Sentinel-2 remote-sensing satellite



Antenna for radio and GPS signals

SMALLER AND CHEAPER

Advances in electronics have made satellites smaller, cheaper, and more robust. Miniature "cubesats" are built from standardized units that slot together, and weigh far less than conventional satellites but have many of the same abilities. They can be quickly built and their small size means they can be launched cheaply, "hitching a lift" on a larger satellite's launch.

SATELLITE ORBITS

Satellites follow different orbits depending on their function. Those that simply need to be above Earth's atmosphere—such as communication satellites (comsats) for satellite phones—use a low Earth orbit (LEO), 120–1,200 miles (200–2,000 km) up. Other comsats often use geostationary orbits, which means that they stay fixed above a point on the equator, while some specialized satellites use highly elliptical (elongated) orbits. Satellites observing Earth use polar orbits, which allow them to cover large areas of the rotating globe in close-up.

